

Git Challenges Documentation

Task 1: Resolve Merge Conflicts

Commands Executed:

- `git checkout -b Conflict`
- `git add .`
- `git commit -m "Modified Sample File"`
- `git checkout main`
- `vi sample`
- `git add .`
- `git commit -m " Modified sample in Main Branch"`
- `git checkout main`
- `git merge conflict`
- `# Resolve conflict`
- `git add .`
- `git commit -m "Resolved merge conflict"`
- `git push origin main`

Required Evidence (Screenshots/Outputs):

```
DELL@KHADEER MINGW64 ~/Desktop/GitTask (main)
$ git checkout conflict
Switched to branch 'conflict'

DELL@KHADEER MINGW64 ~/Desktop/GitTask (conflict)
$ ls
README.md  Samplefile  alpabet  conflict.txt  file1  file2  password

DELL@KHADEER MINGW64 ~/Desktop/GitTask (conflict)
$ vi Samplefile

DELL@KHADEER MINGW64 ~/Desktop/GitTask (conflict)
$ git add Samplefile
warning: in the working copy of 'Samplefile', LF will be replaced by CRLF the next time Git touches it

DELL@KHADEER MINGW64 ~/Desktop/GitTask (conflict)
$ git commit -m "Modified Samplefile in conflict branch"
[conflict 9944aab] Modified Samplefile in conflict branch
1 file changed, 2 insertions(+)
```

```

DELL@KHADEER MINGW64 ~/Desktop/GitTask (conflict)
$ git checkout main
Switched to branch 'main'
Your branch and 'origin/main' have diverged,
and have 1 and 4 different commits each, respectively.
(use "git pull" if you want to integrate the remote branch with yours)

DELL@KHADEER MINGW64 ~/Desktop/GitTask (main)
$ vi Samplefile

DELL@KHADEER MINGW64 ~/Desktop/GitTask (main)
$ git add Samplefile
warning: in the working copy of 'Samplefile', LF will be replaced by CRLF the next time Git touches it

DELL@KHADEER MINGW64 ~/Desktop/GitTask (main)
$ git commit -m "Modified Samplefile in Main branch"
[main e98b7fa] Modified Samplefile in Main branch
1 file changed, 1 insertion(+)

DELL@KHADEER MINGW64 ~/Desktop/GitTask (main)
$ git merge conflict
Auto-merging Samplefile
CONFLICT (content): Merge conflict in Samplefile
Automatic merge failed; fix conflicts and then commit the result.

DELL@KHADEER MINGW64 ~/Desktop/GitTask (main|MERGING)
$ vi Samplefile

```

```

<<<<<< HEAD
Welcome to Main Branch
=====
Welcome to Conflict branch

>>>>>> conflict
~

```

```

Welcome to Conflict branch

```

```

* | 970ee1b Resolved merge conflict
* | 9944aab (conflict) Modified Samplefile in conflict branch
* | 6e5dfda Modified conflict.txt File i conflict branch
* | e98b7fa Modified Samplefile in Main branch
! |
* | ded2bd1 password file added
* | b6ace00 Merge pull request #1 from khadeer329/release
! |
* | 1f13b8b (origin/release, release) Samplefiles Added
! |
* | 5052ded Two files Added
* | b376150 Initial commit

```

Task 2: Recover Deleted Branch

Commands Executed:

- `git checkout -b recovery-branch`
- `git add .`
- `git commit -m "Added recovery file"`
- `git checkout main`
- `git branch -D recovery-branch`
- `git reflog`
- `git checkout -b feature-demo bb312a8`

Required Evidence (Screenshots/Outputs):

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git branch
* main

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git checkout -b recovery-branch
Switched to a new branch 'recovery-branch'

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (recovery-branch)
$ echo "Branch Recovery Demo" > recovery.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (recovery-branch)
$ git add recovery.txt
warning: in the working copy of 'recovery.txt', LF will be replaced by CRLF the next
time Git touches it

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (recovery-branch)
$ git commit -m "Added recovery file"
[recovery-branch 6104f22] Added recovery file
1 file changed, 1 insertion(+)
create mode 100644 recovery.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (recovery-branch)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git branch
* main
  recovery-branch

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git branch -D recovery-branch
Deleted branch recovery-branch (was 6104f22).

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git branch
* main

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git reflog
bb312a8 (HEAD -> main, origin/main, origin/HEAD) HEAD@{0}: checkout: moving from rec
overy-branch to main
6104f22 HEAD@{1}: commit: Added recovery file
bb312a8 (HEAD -> main, origin/main, origin/HEAD) HEAD@{2}: checkout: moving from mai
n to recovery-branch
bb312a8 (HEAD -> main, origin/main, origin/HEAD) HEAD@{3}: clone: from github.com:kh
adeer329/GitChallenge.git

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git checkout -b recovery-branch bb312a8
Switched to a new branch 'recovery-branch'

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (recovery-branch)
$ git branch
  main
* recovery-branch
```

Task 3: Undo Wrong Push

Commands Executed:

- `echo "This is a Wrong Change " > wrong.txt`
- `git add wrong.txt`
- `git commit -m "Added Wrong file"`
- `git push origin main`
- `git revert bb312a8`

Required Evidence (Screenshots/Outputs):

```
517 git checkout main
518 echo "This is a wrong change" > wrong.txt
519 git add wrong.txt
520 git commit -m "Added Wrong file"
521 git push origin main
522 git log --oneline
523 git revert bb312a8
```

```
71ac9b3 Revert "Initial commit"
5cf3866 Added Wrong file
bb312a8 (recovery-branch) Initial commit
```

Task 4: Amend a Commit

Commands Executed:

- `echo "This file 1" > file1.txt`
- `git add file1.txt`
- `git commit -m "Added Fil1"`
- `git add missingfile`
- `echo "This file 2" >> file1.txt`
- `git add file2.txt`
- `git commit -m "Added Fil2"`
- `git commit --amend "Adde File1 And File 2"`

- Required Evidence (Screenshots/Outputs):

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ echo "This is file 1" > file1.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ echo "This is file 2" > file2.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git add file1.txt
warning: in the working copy of 'file1.txt', LF will be replaced by CRLF the next time Git touches it

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git commit -m "Add file1"
[main 88fdd61] Add file1
1 file changed, 1 insertion(+)
create mode 100644 file1.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git commit --amend -m "Add file1 and file2"
[main 5c58d05] Add file1 and file2
Date: Thu Jul 9 15:57:42 2026 +0530
2 files changed, 2 insertions(+)
create mode 100644 file1.txt
create mode 100644 file2.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git log --oneline
5c58d05 (HEAD -> main) Add file1 and file2
412b8af (origin/main, origin/HEAD) Reapply "Added wrong file"
74fb5ae Revert "Added wrong file"
71ac9b3 Revert "Initial commit"
5cf3866 Added wrong file
bb312a8 (recovery-branch) Initial commit
```

Task 5: Cherry-pick a Commit

Commands Executed:

- git checkout feature
- git log --oneline
- git checkout main
- git cherry-pick <commit-hash>
- git log --graph --oneline --all

Required Evidence (Screenshots/Outputs):

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git branch
* main
  recovery-branch

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git checkout recovery-branch
Switched to branch 'recovery-branch'

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (recovery-branch)
$ git log --oneline
bb312a8 (HEAD -> recovery-branch) Initial commit

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (recovery-branch)
$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git branch
* main
  recovery-branch

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git cherry-pick bb312a8
[main cfd3c49] Initial commit
Author: khadeer329 <155939706+khadeer329@users.noreply.github.com>
Date: Thu Jul 9 15:14:08 2026 +0530
2 files changed, 25 insertions(+)
create mode 100644 .gitignore
create mode 100644 README.md

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git log --oneline
cfd3c49 (HEAD -> main) Initial commit
5c58d05 Add file1 and file2
412b8af (origin/main, origin/HEAD) Reapply "Added Wrong file"
74fb5ae Revert "Added Wrong file"
71ac9b3 Revert "Initial commit"
5cf3866 Added Wrong file
bb312a8 (recovery-branch) Initial commit
```

Task 6: Interactive Rebase

Commands Executed:

- echo "Line 1" > demo.txt
- git add demo.txt
- git commit -m "Added demo file "
- echo "Line 2" >> demo.txt
- git add demo.txt
- git commit -m "Updated demo file "
- echo "Line 3" >> demo.txt

- git add demo.txt
- git commit -m "Added third file "
- git rebase -i HEAD~3
- Change pick to squash/reword
- git log --oneline

Required Evidence (Screenshots/Outputs):

```

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git branch
* main
  recovery-branch

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git chekout -b rebase-demo
git: 'chekout' is not a git command. See 'git --help'.

The most similar command is
  checkout

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ echo "Line 1" > demo.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git add demo.txt
warning: in the working copy of 'demo.txt', LF will be replaced by CRLF the next time Git touches it

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git commit -m "Added demo file"
[main 57afc72] Added demo file
1 file changed, 1 insertion(+)
create mode 100644 demo.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ echo "Line 2" >> demo.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git add demo.txt
warning: in the working copy of 'demo.txt', LF will be replaced by CRLF the next time Git touches it

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git commit -m "update demo file"
[main bb6ceb5] update demo file
1 file changed, 1 insertion(+)

```

```

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ echo "Line 3" >> demo.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git commit -m "Added third file"
On branch main
Your branch is ahead of 'origin/main' by 4 commits.
(use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   demo.txt

no changes added to commit (use "git add" and/or "git commit -a")

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git add demo.txt
warning: in the working copy of 'demo.txt', LF will be replaced by CRLF the next time Git touches it

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git commit -m "Added third file"
[main 361e58a] Added third file
1 file changed, 1 insertion(+)

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git log --oneline
361e58a (HEAD -> main) Added third file
bb6ceb5 update demo file
57afc72 Added demo file
cfd3c49 Initial commit
5c58d05 Add file1 and file2
412b8af (origin/main, origin/HEAD) Reapply "Added Wrong file"
74fb5ae Revert "Added Wrong file"
71ac9b3 Revert "Initial commit"
5cf3866 Added Wrong file
bb312a8 (recovery-branch) Initial commit

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git log --oneline
8123644 (HEAD -> main) Added Demo file with all updates
cfd3c49 Initial commit
5c58d05 Add file1 and file2
412b8af (origin/main, origin/HEAD) Reapply "Added Wrong file"
74fb5ae Revert "Added Wrong file"
71ac9b3 Revert "Initial commit"
5cf3866 Added Wrong file
bb312a8 (recovery-branch) Initial commit

```

Task 7: Tagging & Release

Commands Executed:

- git tag v1.0
- git push origin v1.0
- git tag -d v1.0
- git push origin --delete v1.0
- git tag v1.0 <commit>

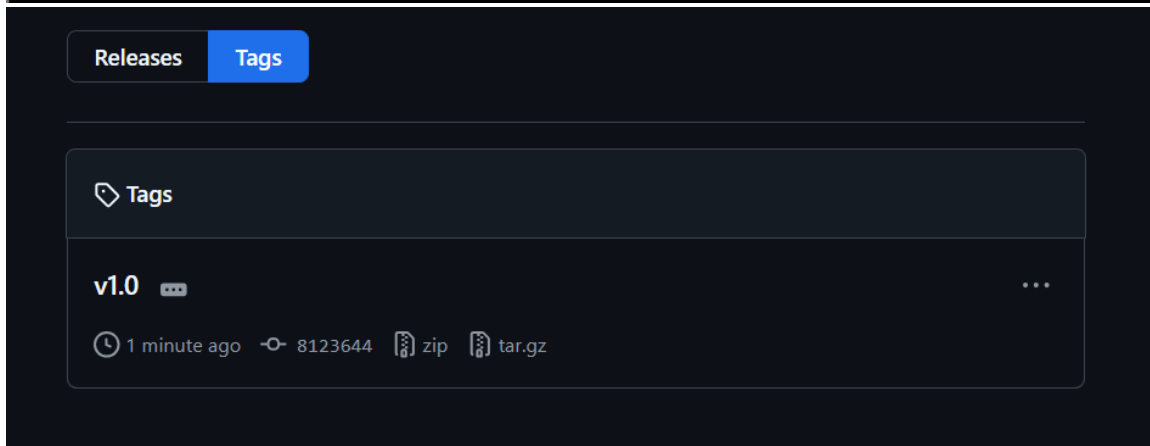
- `git push origin v1.0`

Required Evidence (Screenshots/Outputs):

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git tag -a v1.0 -m "Version 1.0 Release"

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git tag
v1.0

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git push origin v1.0
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 8 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (12/12), 1.21 KiB | 617.00 KiB/s, done.
Total 12 (delta 2), reused 2 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To github.com:khadeer329/GitChallenge.git
 * [new tag]          v1.0 -> v1.0
```



```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git tag -d v1.0
Deleted tag 'v1.0' (was 16c1296)

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git tag

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git push origin --delete v1.0
To github.com:khadeer329/GitChallenge.git
- [deleted]          v1.0
```

Releases

Tags



There aren't any releases here

You can create a release to package software, along with release notes and links to binary files, for other people to use. Learn more about releases in [our docs](#).

Create a new release

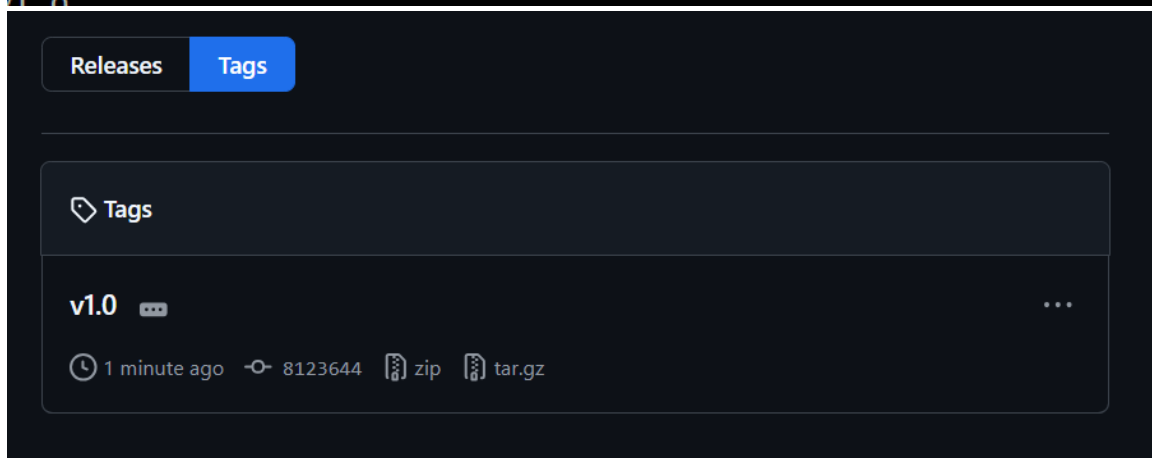
```

ELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git tag -fa v1.0 8123644 -m "Version 1.0 Release"
Updated tag 'v1.0' (was 8123644)

ELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git push origin v1.0
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 8 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (12/12), 1.21 KiB | 617.00 KiB/s, done.
Total 12 (delta 2), reused 2 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To github.com:khadeer329/GitChallenge.git
 * [new tag]          v1.0 -> v1.0

ELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git tag
v1.0

```



Task 8: Clone with Sparse Checkout

Commands Executed:

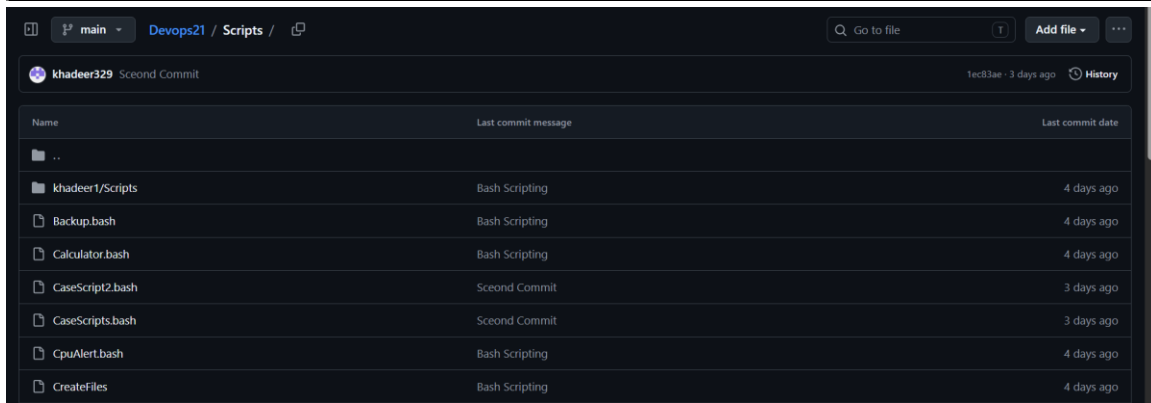
- `git clone --filter=blob:none --sparse <repo-url>`
- `cd <repo>`
- `git sparse-checkout set <subdirectory>`

Required Evidence (Screenshots/Outputs):

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ git clone --no-checkout git@github.com:khadeer329/Devops21.git
Cloning into 'Devops21'...
remote: Enumerating objects: 35, done.
remote: Counting objects: 100% (35/35), done.
remote: Compressing objects: 100% (32/32), done.
remote: Total 35 (delta 1), reused 34 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (35/35), 5.32 KiB | 1.33 MiB/s, done.
Resolving deltas: 100% (1/1), done.

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge (main)
$ cd Devops21

DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge/Devops21 (main)
$ git sparse-checkout init --cone
```



```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/GitChallenge/Devops21/Devops21 (main|SPARSE)
$ ls
Scripts/
```

Task 9: Reset vs Revert

Commands Executed:

- `git reset --hard HEAD~1`
- `git revert HEAD`

Required Evidence (Screenshots/Outputs):

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ echo "Version 1" > demo.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ git add demo.txt
warning: in the working copy of 'demo.txt', LF will be replaced by CRLF the next time Git touches it

DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ git commit -m "Initial commit"
[master (root-commit) 976b082] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 demo.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ echo "Version 2" >> demo.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ git add demo.txt
warning: in the working copy of 'demo.txt', LF will be replaced by CRLF the next time Git touches it

DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ git commit -m "Second commit"
[master c6fe79d] Second commit
1 file changed, 1 insertion(+)

DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ echo "Version 3" >> demo.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ git add demo.txt
warning: in the working copy of 'demo.txt', LF will be replaced by CRLF the next time Git touches it

DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ git commit -m "Third commit"
[master fd3a014] Third commit
1 file changed, 1 insertion(+)
```

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ git log --oneline
fd3a014 (HEAD -> master) Third commit
c6fe79d Second commit
976b082 Initial commit
```

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ git reset --hard HEAD~1
HEAD is now at c6fe79d Second commit

DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ git log --oneline
c6fe79d (HEAD -> master) Second commit
976b082 Initial commit
```

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ git log --oneline
77f2b5d (HEAD -> master) Revert "Third commit"
6afc347 Third commit
c6fe79d Second commit
976b082 Initial commit
```

Task 10: Detached HEAD

Commands Executed:

- `git checkout <commit-hash>`
- `git switch -c new-branch`

Required Evidence (Screenshots/Outputs):

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ git log --oneline
1bd3913 (HEAD -> master) "Second commit"
c6fe79d Second commit
976b082 Initial commit

DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ git checkout 1bd3913
Note: switching to '1bd3913'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

    git switch -c <new-branch-name>

Or undo this operation with:

    git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 1bd3913 "Second commit"

DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo ((1bd3913...))
$ git status
HEAD detached at 1bd3913
nothing to commit, working tree clean

DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo ((1bd3913...))
$ git branch
* (HEAD detached at 1bd3913)
  master
```

Task 11: Git Hooks

Commands Executed:

- Create `.git/hooks/pre-commit`
- `chmod +x .git/hooks/pre-commit`
- `git commit -m "Invalid"`
- `git commit -m "JIRA-101 Valid"`

Required Evidence (Screenshots/Outputs):

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ echo "Test" > test.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ git add test.txt
warning: in the working copy of 'test.txt', LF will be replaced by CRLF the next time Git touches it

DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ git commit -m "Added test file"
ERROR: Commit message must start with JIRA-⟨number⟩:
Example: JIRA-101: Add login page

DELL@KHADEER MINGW64 ~/Desktop/khadeer/reset-revert-demo (master)
$ git commit -m "JIRA-101: Added test file"
[master 9c42a22] JIRA-101: Added test file
1 file changed, 1 insertion(+)
create mode 100644 test.txt
```

```
#!/bin/sh

commit_msg_file=$1
commit_msg=$(cat "$commit_msg_file")

if ! echo "$commit_msg" | grep -Eq '^JIRA-[0-9]+:'; then
    echo "ERROR: Commit message must start with JIRA-⟨number⟩:"
    echo "Example: JIRA-101: Add login page"
    exit 1
fi

exit 0

~
```

Task 12: Squash Merge vs Rebase Merge

Commands Executed:

- `git merge --squash feature`
- `git rebase feature`

Required Evidence (Screenshots/Outputs):

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (master)
$ echo "Main Branch" > file.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (master)
$ git add .
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (master)
$ git commit -m "Initial commit"
[master (root-commit) 5417b8e] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 file.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (master)
$ git log --oneline
5417b8e (HEAD -> master) Initial commit
```

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (master)
$ git checkout -b feature
Switched to a new branch 'feature'

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (feature)
$ echo "Feature Line 1" >> file.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (feature)
$ git add .
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (feature)
$ git commit -m "Feature Commit 1"
[feature fa8d87e] Feature Commit 1
1 file changed, 1 insertion(+)
```

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (feature)
$ echo "Feature Line 2" >> file.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (feature)
$ git add .
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (feature)
$ git commit -m "Feature Commit 2"
[feature 418a2cb] Feature Commit 2
1 file changed, 1 insertion(+)
```

```

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (feature)
$ echo "Feature Line 3" >> file.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (feature)
$ git add .
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (feature)
$ git commit -m "Feature Commit 3"
[feature 8ec0f9a] Feature Commit 3
1 file changed, 1 insertion(+)

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (feature)
$ git log --oneline --graph --all
* 8ec0f9a (HEAD -> feature) Feature Commit 3
* 418a2cb Feature Commit 2
* fa8d87e Feature Commit 1
* 5417b8e (master) Initial commit

```

```

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (master)
$ git merge --squash feature
Updating 5417b8e..8ec0f9a
Fast-forward
Squash commit -- not updating HEAD
file.txt | 3 +++
1 file changed, 3 insertions(+)

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (master)
$ git commit -m "Squash merge feature branch"
[master c52bdbf] Squash merge feature branch
1 file changed, 3 insertions(+)

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (master)
$ git log --oneline --graph --all
* c52bdbf (HEAD -> master) Squash merge feature branch
| * 8ec0f9a (feature) Feature Commit 3
| * 418a2cb Feature Commit 2
| * fa8d87e Feature Commit 1
| /
* 5417b8e Initial commit

```

Rebase:

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (master)
$ git checkout -b feature-rebase
Switched to a new branch 'feature-rebase'

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (feature-rebase)
$ echo "Rebase Line 1" >> file.txt
git add .
git commit -m "Rebase Commit 1"

echo "Rebase Line 2" >> file.txt
git add .
git commit -m "Rebase Commit 2"

echo "Rebase Line 3" >> file.txt
git add .
git commit -m "Rebase Commit 3"
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it
[feature-rebase b389bfc] Rebase Commit 1
1 file changed, 1 insertion(+)
create mode 100644 file.txt
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it
[feature-rebase 053c163] Rebase Commit 2
1 file changed, 1 insertion(+)
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it
[feature-rebase e0ecd19] Rebase Commit 3
1 file changed, 1 insertion(+)

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (feature-rebase)
$ git checkout main
error: pathspec 'main' did not match any file(s) known to git

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (feature-rebase)
$ git checkout master
Switched to branch 'master'

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (master)
$ git checkout feature-rebase
Switched to branch 'feature-rebase'

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (feature-rebase)
$ git rebase master
Current branch feature-rebase is up to date.

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (feature-rebase)
$ git checkout master
Switched to branch 'master'

DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (master)
$ git merge feature-rebase
Updating cb57e9c..e0ecd19
Fast-forward
 file.txt | 3 +++
1 file changed, 3 insertions(+)
create mode 100644 file.txt

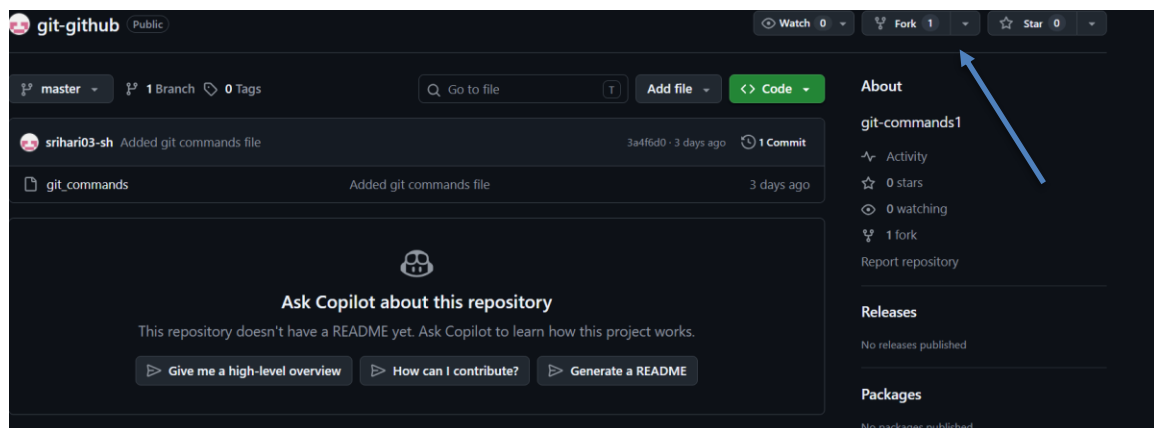
DELL@KHADEER MINGW64 ~/Desktop/khadeer/merge-demo (master)
$ git log --oneline --graph --all
* e0ecd19 (HEAD -> master, feature-rebase) Rebase Commit 3
* 053c163 Rebase Commit 2
* b389bfc Rebase Commit 1
* cb57e9c Revert "Initial commit"
| * 8ec0f9a (feature) Feature Commit 2
| * 418a2cb Feature Commit 2
| * fa8d87e Feature Commit 1
|/
* 5417b8e Initial commit
```

Task 13: Fork & Pull Request Workflow

Commands Executed:

- Fork repo
- `git clone git@github.com:khadeer329/git-github.git`
- `git checkout -b feature-update`
- `touch README.md`
- `git commit -m "Added README file"`
- `git push origin feature-update`
- Create Pull Request

Required Evidence (Screenshots/Outputs):



```
DELL@KHADEER MINGW64 ~/Desktop/khadeer
$ git clone git@github.com:srihari03-sh/git-github.git
Cloning into 'git-github'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
```

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer
$ cd git
GitChallange/ GitTask/      git-github/
```

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer
$ cd git-github/
```

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/git-github (master)
$ git checkout -b feature-update
Switched to a new branch 'feature-update'
```

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/git-github (feature-update)
$ git branch
* feature-update
  master
```

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/git-github (master)
$ git checkout -b feature-update
Switched to a new branch 'feature-update'
```

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/git-github (feature-update)
$ git branch
* feature-update
  master
```

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/git-github (feature-update)
$ touch README.md
```

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/git-github (feature-update)
$ vi README.md
```

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/git-github (feature-update)
$ git add .
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
```

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/git-github (feature-update)
$ git commit -m "Added And Updated README with assignment changes"
[feature-update 32adcd0] Added And Updated README with assignment changes
1 file changed, 1 insertion(+)
create mode 100644 README.md
```

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/git-github (feature-update)
$ git log --oneline
32adcd0 (HEAD -> feature-update) Added And Updated README with assignment changes
3a4f6d0 (origin/master, origin/HEAD, master) Added git commands file
```

Comparing changes

Choose two branches to see what's changed or to start a new pull request. If you need to, you can also [compare across forks](#) or [learn more about diff comparisons](#).

 base repository: srihari03-sh/git-github

base: master

...

head repository: khadeer329/git-github

compare: feature-update

✓ **Able to merge.** These branches can be automatically merged.



Add a title *

Added And Updated README with assignment changes

Helpful resources

[GitHub Community Guidelines](#)

Add a description

Write

Preview

H B I    |    |     

Add your description here...

 Markdown is supported

 Paste, drop, or click to add files

☒ Allow edits by maintainers  Create pull request

The top screenshot shows a GitHub pull request titled "Added And Updated README with assignment changes #1". The pull request is from the branch "khadeer329:feature-update" to the "srihari03-sh:master" branch. It includes a comment from "khadeer329" stating "No description provided." and a commit history showing "Added And Updated README with assignment changes" by "32adcd8". A green checkmark indicates "No conflicts with base branch". The bottom screenshot shows the repository page for "git-github" by "srihari03-sh". It displays the README section with an "Add a README" button and repository statistics including 2 branches, 0 tags, 0 stars, 0 watching, and 0 forks.

Task 14: Recover Lost Commit

Commands Executed:

- `git reset --hard HEAD~1`
- `git reflog`
- `git reset --hard b659f18`

Required Evidence (Screenshots/Outputs):

```
DELL@KHADEER MINGW64 ~/Desktop/khadeer/recover-commit-demo (master)
$ echo "This commit will be recovered" >> demo.txt

DELL@KHADEER MINGW64 ~/Desktop/khadeer/recover-commit-demo (master)
$ git add .
warning: in the working copy of 'demo.txt', LF will be replaced by CRLF the next time Git touches it

DELL@KHADEER MINGW64 ~/Desktop/khadeer/recover-commit-demo (master)
$ git commit -m "Recover me"
[master b659f18] Recover me
1 file changed, 1 insertion(+)

DELL@KHADEER MINGW64 ~/Desktop/khadeer/recover-commit-demo (master)
$ git commit -m "Recover me"
On branch master
nothing to commit, working tree clean

DELL@KHADEER MINGW64 ~/Desktop/khadeer/recover-commit-demo (master)
$ git log --oneline
b659f18 (HEAD -> master) Recover me
4dc2b03 Initial commit

DELL@KHADEER MINGW64 ~/Desktop/khadeer/recover-commit-demo (master)
$ git reset --hard HEAD~1
HEAD is now at 4dc2b03 Initial commit

DELL@KHADEER MINGW64 ~/Desktop/khadeer/recover-commit-demo (master)
$ git log --oneline
4dc2b03 (HEAD -> master) Initial commit

DELL@KHADEER MINGW64 ~/Desktop/khadeer/recover-commit-demo (master)
$ git reflog
4dc2b03 (HEAD -> master) HEAD@{0}: reset: moving to HEAD~1
b659f18 HEAD@{1}: commit: Recover me
4dc2b03 (HEAD -> master) HEAD@{2}: commit (initial): Initial commit

DELL@KHADEER MINGW64 ~/Desktop/khadeer/recover-commit-demo (master)
$ git reset --hard b659f18
HEAD is now at b659f18 Recover me

DELL@KHADEER MINGW64 ~/Desktop/khadeer/recover-commit-demo (master)
$ git log --oneline
b659f18 (HEAD -> master) Recover me
4dc2b03 Initial commit
```